# Jayati Ghori

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## EDUCATION

**Master of Science (MS) – Computer Science** 08/2025 – 05/2027

Arizona State University, AZ | GPA: 3.78

**Coursework:** Cloud Computing, Advanced Data Structures & Algorithms, Data Processing at Scale, Fundamentals of Software Engineering

**Bachelor of Technology (B.Tech) – Information Technology** 03/2020 – 05/2024

Dharmsinh Desai University, India | GPA: 3.67

**Coursework:** Data Analysis & Information Extraction, Knowledge Systems, Database Management Systems, Data Structures & Algorithms

## TECHNICAL SKILLS

**Languages:** Python, Java, C++, SQL, HTML, CSS, jQuery, Bootstrap, JavaScript
**AI / Machine Learning:** PyTorch, scikit-learn, pandas, NumPy, FastAPI, Computer Vision (MTCNN, FaceNet), NLP fundamentals
**Cloud & Data Engineering:** AWS (EC2, S3, SQS, Lambda, IoT Greengrass), Apache Spark (Core, SQL, Streaming), Hadoop/MapReduce, Docker, PostgreSQL, MongoDB
**Software Engineering:** System Design, Low-Latency Systems Design, Algorithm Design & Complexity Analysis, Distributed Systems, CMake

## PROJECTS

**AI Size & Fit Recommender** | Machine Learning, Python 08/2026 – 09/2026

- Improved clothing **size-fit prediction** by **+79.7% macro-F1 over baseline (0.5088)** using a **Random Forest** classifier (**scikit-learn**) trained on the **RentTheRunway dataset** (192K transactions, 105K users, 5.8K items) to reduce e-commerce returns.
- Engineered 18 **feature-engineered** inputs, including a custom **size_delta** metric, and used **GroupShuffleSplit** to guarantee zero train/test leakage across users.
- Handled **model deployment** via a FastAPI backend and a live **Streamlit Cloud** app, building an **inference pipeline** that loads the model from cloud storage at runtime and returns ranked size predictions with confidence scores.

**Elastic Face Recognition Pipeline (IaaS → Serverless → Edge)** | Python, AWS, PyTorch 08/2025 – 12/2025

- Built a 3-tier **computer vision** architecture directly on **AWS IaaS** primitives (**EC2, S3, SQS**) to run **model inference** on image uploads, scoring **100/100** with a custom **autoscaling** controller (0–15 EC2 instances) driven by live queue depth.
- Migrated to a **serverless FaaS** inference pipeline using **AWS Lambda** and **MTCNN/FaceNet** (**PyTorch**), achieving **100% successful completions** and **sub-3-second average latency** (95/100).
- Extended the pipeline to **edge inference** with **AWS IoT Greengrass** and **MQTT**, running detection locally while offloading heavier **model serving** workloads to the cloud (90/100).

**Limit Order Book (LOB) Matching Engine** | C++20, High-Performance Systems 07/2026 – 08/2026

- Engineered a low-latency matching engine achieving ∼**9.56M orders/sec** throughput with **true O(1) order cancellation** (350–440ns flat latency), the same **cache-optimized**, zero-allocation systems design that underpins low-latency **ML model-serving** and **inference** infrastructure.
- Designed a custom **memory pool** with an intrusive free-list, cutting allocation latency **21.2x** versus **std::list**, and migrated to cache-optimized data structures, techniques directly transferable to minimizing **latency jitter** in high-throughput inference-serving systems, validated across a 37-case **ASan/UBSan** test suite.

## WORK EXPERIENCE

**Office Assistant – The Polytechnic School Director's Office, ASU** 02/2026 – Present

- Manage **front-desk operations** and serve as the primary point of contact for visitors, calls, and mail, applying strong **professional communication** and organizational skills in a fast-paced administrative environment.

**Software Engineering Intern – Escrow Infotech, India** 12/2023 – 04/2024

- Engineered a functional Android prototype (**BabyCam**) end-to-end in **Kotlin** using **MVVM architecture**, solving real-time device-to-device video streaming over **Wi-Fi Direct** with no internet or access point required.
- Collaborated with the engineering team on **software testing** and code review, applying **Agile** practices to ship a maintainable, well-tested prototype.

## CERTIFICATIONS

- **Artificial Intelligence Course – Rinex Organization / Skill India (NSDC), 2023: Machine Learning**, **Deep Learning**, **Computer Vision**, **NLP**
- **AWS Academy Graduate – Machine Learning Foundations:** Amazon SageMaker, Rekognition, NLP on AWS, ML pipeline implementation